



Prefixoids ≠ prefixes: not all inseparable complex verbs are (underlyingly) the same

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Abstract

The goal of this paper is to investigate the structural distinctions that exist between *prefixoids* and *prefixes* in modern Dutch, with an empirical focus on *inseparable complex verbs* (ICVs). Adopting a derivational, separationist approach to the syntax-morphology interface (*Distributed Morphology*), we maintain that the fundamental difference between *prefixoids* and *prefixes* is that the former are clitics while the latter are initially merged with $\sqrt{\text{roots}}$ lower in the first phase (vP) prior to when these complex predicates are labeled (by v). As clitics, prefixoids are adjoined to the left-edge of the first phase (with the phase head v). Our theoretical claims are further substantiated by data extracted from the Dutch *Celex Lexicon*, providing a detailed assessment of the categorical status of the stem that prefixoids and prefixes attach to, their ability to contribute a resultative interpretation, and their potential effects on the argument structure of the ICV.

Keywords Prefixoid-prefix distinction · Complex predicates · Inseparable complex verbs · Dutch · Distributed Morphology

1 Introduction

The aim of this paper is to investigate the main structural differences between *prefixes* and *prefixoids* in Dutch. To achieve this goal we take a closer look at the inventory of pre-verbal elements in Dutch, and advance the claim that beyond the standard division between **separable complex verbs** (SCVs) and **inseparable complex verbs** (ICVs) (see, e.g., Booij 1990; Roberts 1991; Neeleman 1994; Koopman 1995; Wurmbrand 1998, 2000; Zeller 2001; McIntyre 2002; Ramchand and Svenonius 2002; Biskup et al. 2011), the latter category consists of at least two distinct sub-classes, namely, (i) **prefix-headed**-ICVs (prefix-ICVs for short) and (ii) **prefixoid-headed**-ICVs (prefixoid-ICVs

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for short). A canonical example of a SCV is given in (1), of a prefix-ICV in (2), and of a prefixoid-ICV in (3). For the remainder of this paper, we focus on the distinction between prefix-ICVs (2) and prefixoid-ICVs (3).

- (1) De vrouw **leest** het boek *door*.
the woman reads the book through
 ‘The woman reads through the book.’ [SCV]
- (2) De vrouw **ver-koopt** haar huis.
the woman PREF-buy her house
 ‘The woman sells her house.’ [prefix-ICV]
- (3) De vrouw **achter-haalt** de waarheid.
the woman behind-fetch the truth
 ‘The woman recovers the truth.’ [prefixoid-ICV]

The key morphosyntactic difference between SCVs on the one hand and both classes of ICVs on the other, is that in the former, the particle is separated (hence *seperable* complex verb) from the verbal part of the complex verb in V2-constructions (1),¹ whereas prefixes and prefixoids appear together, i.e., are bound, with the verbal part in V2-position (2)–(3). The fundamental morphosyntactic distinction between prefixes and prefixoids in these ICVs is the fact that prefixes cannot be used as free morphemes, whereas prefixoids can (e.g., *achter* ‘behind’ can also be used as a standalone preposition). Given that they can appear as free morphemes, but seem to be used as a sort of prefix in ICVs, these elements have occasionally been labeled as *prefixoids* in the literature on Dutch morphology (Smessaert et al. 2019). The combination of a prefixoid and verb has also been called a *pseudocompound* by the same authors, due to the fact that these prefixoid-ICVs comprise two free morphemes and may thus be considered compounds, but at the same time they do not behave as canonical compounds. For example, in Dutch verbal compounds—a rather unproductive class of compounds, see, e.g., de Haas and Trommelen (1993), Ackema (1999), Smessaert et al. (2019)—the main stress always falls on the left-hand member of the compound, like canonical compounds in the language. In contrast, the main stress in prefixoid- and prefix-ICVs falls on the head of the complex verb.

The identical stress pattern, nonseparability, and other shared features between prefix-ICVs and prefixoid-ICVs have led authors of earlier studies (e.g., Zwanenburg 1997 and Blom 2005) to argue for a unified analysis of these ICVs as complex verbs with a prefixed element. However, these previous studies did not carry out a detailed investigation into the morphosyntactic and semantic properties of prefix-ICVs and prefixoid-ICVs, which we will do in this paper.

Two important empirical observations we make straight away are related to the prefixoid/prefix category and the (im)possibility of stacking these two pre-elements in a sequence at the left-edge of the complex predicate. With respect to the category

¹ Besides in V2-constructions, the particle and verb of SCVs are also separated by the infinitival marker *te* ‘to’ (e.g., *door te lezen* lit. through to read ‘to be read through’), by the prefix-part of the perfect participle circumfix (e.g., *door-ge-lez-en* ‘read through’), and can be separated from the verb by auxiliary/modal verbs in verb clusters (e.g., *door moeten lezen* lit. through must read ‘having to read (it) through’). None of these separation scenarios occur with ICVs.

of the prefixoid/prefix, it is noteworthy that diachronically all Dutch verbal prefixes stem from prepositions (de Haas and Trommelen 1993), whereas the prefixoids are derived from three categories, namely (i) prepositions, (ii) adverbs, and (iii) adjectives. An example of the first category was already presented in (3) above; examples of the second and third categories are given in (4) and (5), respectively.

- (4) Het geluid **weer-galmt** hier indrukwekkend.
the sound again-echo here impressively
 ‘The sounds re-echos here impressively.’ [adv-headed prefixoid]
- (5) Hij **vol-doet** niet aan de eisen.
he full-does not at the requirements
 ‘He doesn’t meet the requirements.’ [adj-headed prefixoid]

An empirical observation that has, to the best of our knowledge, never played a significant role in the discussion of prefix(oids) in Dutch and related languages concerns the (in)ability to stack prefix- and prefixoid-elements in complex verbs. While the stacking of two prefixes (6) and two prefixoids (7) results in ungrammatical structures, prefixoid-prefix-‘stacked’ complex structures are well-formed (8-a). Interestingly, the opposite stacked order, i.e. prefix-prefixoid, is impossible (8-b).

- (6) *Ver-be-licht-en.
PREF-PREF-light-INF
- (7) *Over-achter-hal-en.
over-behind-take-INF
- (8) a. over-be-licht-en
over-PREF-light-INF
 ‘overexpose’
 b. *be-over-licht-en
PREF-over-light-INF

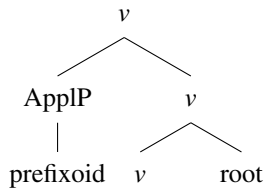
These stacking data are important for two reasons: First, they show that prefixoids and prefixes cannot be assumed to occupy exactly the same position in the (functional sequence of the) *v*/VP, otherwise all stacking data would be ungrammatical. Second, the grammaticality of prefixoid-prefix stacking in (8-a) indicates that prefixoids attach to the verbal stem *after* genuine prefixes have been added/derived.² We will elaborate on these data and how they can be accounted for by our analysis of the prefix-prefixoid distinction throughout the remainder of the paper.

We build our analysis of the structural distinction that exists between prefixes and prefixoids in ICVs on three theory-independent criteria: namely, whether they can (i) ‘verbalize’ (e.g., combine with) a nonverbal stem to form an ICV, (ii) add resultativity, and (iii) have an effect on the argument structure of the stem they attach to. In what follows, we demonstrate that prefix-ICVs and prefixoid-ICVs behave differently when it comes to these three criteria and should therefore not receive the same analysis (see

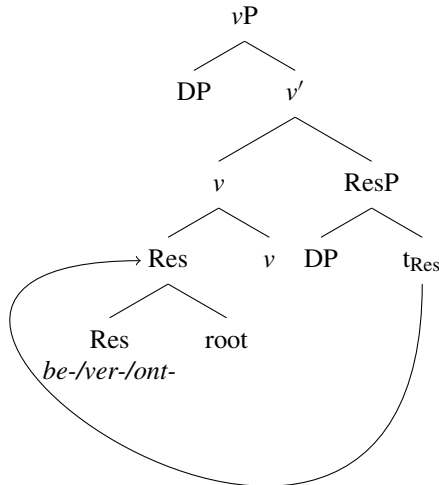
² We expand upon this in our analysis in Sect. 3; note that it is a well-established generalization that the merging of elements within a phase and those that adjoin to the phase edge belong to separate Spell-Out domains (Uriagereka 1999).

Sect. 3.2). We classify prefixoids as clitics, which adjoin to the v phase head, whereas prefixes lexicalize a Res-head, to which the root of the ICV adjoins. As such, prefixoids and prefixes differ fundamentally from one another in terms of (i) how they enter the narrow syntax and (ii) their position in the first phase (i.e., the vP , the phase in which the event is built (Ramchand 2008). It is on the basis of these distinct structures that we build our analysis of prefixoid-ICVs and prefix-ICVs (see (9) and (10), respectively).

(9) **Structure of prefixoid-ICVs**



(10) **Structure of prefix-ICVs**



This paper is structured as follows: In Sect. 2, we provide a detailed empirical overview of Dutch ICVs. We annotated data extracted from the *Dutch Celex Lexicon* based on the three aforementioned criteria for prefixoid-ICVs ($n = 195$) and prefix-ICVs ($n = 400$). In general, the results of the annotation study show that prefixes have more impact on the syntax and semantics of the first phase compared to prefixoids; i.e., they show higher frequencies for all three properties. Furthermore, the results show a difference between P-based prefixoids and A(dv)-based prefixoids, in that the former still seem to impact the first phase (though to a lesser degree than prefixes), whereas the latter have virtually no impact on the syntax and semantics of the first phase. We integrate these empirical findings into a theoretical analysis in Sect. 3. Adopting the primary tenets of *Distributed Morphology* (Halle and Marantz 1993; Embick and Noyer 2007), we propose that the fundamental difference between prefixoids and prefixes can be

reduced to the clitic status of the former (relying on Pair Merge to the phase head v), while prefixes are merged in narrow syntax prior to the completion of the first phase. Building on Folli and Harley's (2020) analysis of bimorphemic English verbs, we also argue that prefixes lexicalize Res, to which the root of the ICV is then adjoined, i.e., prior to verbalization. Section 4 concludes the paper.

2 Properties of Dutch prefixoid- and prefix-ICVs

In order to discover whether there are differences between prefixoids and prefixes besides the order in which they can stack, and prefixoids' capacity to be used as standalone elements, we carried out an in-depth annotation study into several important properties of prefixoids and prefixes. We focused on properties which might reveal the degree to which prefixoids and prefixes have a syntactic or semantic effect on the first phase. As already mentioned in the introduction, we investigated the following three properties:

- **Property I:** whether or not prefixoids/prefixes combine with a nonverbal stem to form an ICV
- **Property II:** whether or not prefixoids/prefixes have a resultative effect on the compositional semantics of the ICV they are part of
- **Property III:** whether or not prefixoids/prefixes have an effect on the argument structure of the stem they attach to

In Sect. 2.1, we discuss the methodology that was used. In Sect. 2.2, we present the results of the annotation study.

2.1 Methodology

In this section, we describe how we generated the samples of prefixoid-ICVs and the prefix-ICVs. We furthermore make explicit for which syntactic and semantic properties these samples were annotated and how the annotation was executed.

2.1.1 Creation of samples

Based on a full list of prefixoids which occur in ICVs as found in the *Algemene Nederlandse Spraakkunst* 'General Dutch Language' (Haeseryn et al. 1997), we searched through the *Dutch Celex Lexicon* (Baayen et al. 1995) for all instances of Dutch prefixoid-ICVs. The total number of prefixoid-ICVs (unique word forms) was 195. Annotating these forms manually was a feasible task, and all prefixoid-ICVs were thus included for annotation. A random sample of prefix-ICVs was selected from the same corpus focusing on the three verbalizing prefixes in Dutch, namely *ver-*, *ont-*, and *be-*. For *ver-* we made a random selection of 150 verbs out of a total of 618 unique word forms, for *ver-* 150 verbs out of a total of 358 unique word forms, and for *ont-* 100 verbs out of a total of 192 unique word forms. The combined sample of the three types of prefix-ICVs is a total of 400 prefix-ICVs.

2.1.2 Annotation of properties

Before starting the actual annotation, we first annotated the category of the prefixoid in all the prefixoid-ICV datapoints. As already mentioned in the introduction, prefixoids in Dutch prefixoid-ICVs can be prepositions, adverbs or adjectives (Haeseryn et al. 1997). To date, however, the exact frequencies of these categories in prefixoid-ICVs has never been investigated. All annotations were done by the first author, who is a native speaker of Dutch.

Both the prefixoid-ICV and the prefix-ICV data set was further annotated for the three properties presented at the beginning of this section. Regarding Property I (whether the prefixoid/prefix can attach to a nonverbal stem), it is known that the Dutch prefixes under consideration are able to combine with a nonverbal stem (de Haas and Trommelen 1993). As such, these three prefixes can be said to have ‘verbalizing capacity’, i.e., are able to take a nonverbal stem and create a verbal output (see, e.g., Vanden Wyngaerd et al. (2022) for a Nanosyntactic account of Dutch deadjectival *ver*-ICVs, but see Neeleman and Schipper (1993) for a different analysis of such verbs, in which an empty verbalizing head intervenes between the nonverbal stem and the prefix). Consider the following examples of *ver*-ICVs:

- (11) a. plicht
 duty
 ‘duty’
 b. ver-plicht-en
 PREF-duty-INF
 ‘to oblige’
- (12) a. blind
 blind
 ‘blind’
 b. ver-blind-en
 PREF-blind-INF
 ‘to make blind’
- (13) a. *liez-
 non-lexical-base
 b. ver-liez-en
 PREF-liez-INF
 ‘to lose’

In (11), *ver-* combines with a nominal stem, in (12) with an adjectival stem, and in (13) with a non-lexical stem (henceforth NLS), i.e., with a stem that cannot appear without a categorizing affix. Apart from these three types of stems, the three prefixes under consideration can also combine with a verbal stem, an example of which is given in (14).

- (14) a. draai-en
 turn-INF
 ‘to turn’

- b. ver-draai-en
PREF-turn-INF
 ‘to twist’

Until now, no investigation has been carried out addressing whether Dutch prefixoids in prefixoid-ICVs are also capable of attaching to non-verbal stems, like the examples of *ver-* in (11)–(13). Properties II and III were chosen for the same reason; i.e., both a resultative effect and an effect on the argument structure of the stem have been reported as core properties of Dutch verbalizing prefixes (Hoekstra et al. 1987), but these properties have not yet been systematically investigated for prefixoid-ICVs. All three properties additionally relate to the syntactic structure and semantics of the first phase and will thus be very informative for the theoretical analysis of Dutch prefixoid-ICVs and prefix-ICVs. For each annotation, the author consulted the *Van Dale* Dutch dictionary online (professional version), as well as other native speakers and the etymological dictionary when necessary. The data set has been made Open Access via the Open Science Framework, and can be found [here](#).

We now turn to presenting how we annotated the data for the three properties under consideration. We first discuss the annotation of Property I, i.e., the category of the stem to which the prefixoids/prefixes attach. For each ICV, the stem was identified and its category was labeled as either V (verbal), N (nominal), A (adjectival), or NLS (non-lexical stem), with each categorization checked against the *Van Dale* Dutch dictionary. Stems were categorized as N or A only when they were unambiguously nominal or adjectival. In cases of verbal vs. nominal/adjectival ambiguity, we conservatively assigned these stems to the verbal category. For example, in the prefix-ICV *ver-rooken* (lit.PREF-smoke-INF), the stem may be analysed as either of category V or N; in such cases we label it as V.

A final note with respect to Property I concerns stems which contain other derivational morphology. For example, the prefix-ICV *ver-fromm-el-en* lit. PREF-NLS-SUFF-EN contains the verbalizing suffix *-el*. Even though this suffix attaches to a non-lexical stem, this stem combined with verbalizing suffix *-el* forms the stem to which the prefix attaches, and hence that stem was labeled as V.

Second, we discuss the annotation of Property II, i.e., the resultative effect of the prefixoid/prefix on the compositional semantics of the ICV. In order to test this, we applied Wechsler’s test (2005) for resultativity both on the stem of the ICV in cases of a verbal stem, and on the ICV itself. We did this to find out whether the attachment of the prefixoid/prefix has a resultative effect. Let us first illustrate Wechsler’s test for resultativity, which is shown in (15) (Wechsler 2005, 256).

- (15) Resultative (predicate in *italics*; its subject underlined):
 John hammered the metal *flat*.
 ⇒ ‘John hammered the metal; as a result, the metal became flat.’

The sentence in (15) and its entailment show that we are dealing with a resultative predicate (‘flat’), expressing the final (result) state of the sentential object referent, namely ‘the metal’. As previously mentioned, it has been claimed that the Dutch prefixes *be-*, *ver-*, and *ont-* have a resultative effect (Hoekstra et al. 1987). To test this systematically and to extend the test to the potential resultative effect of prefixoids

on the compositional semantics of the ICV, we applied Wechsler's test to both the verbal stem and the ICV. As an example, to find out whether the prefix *be-* contributes resultative meaning to the prefix-ICV *belopen* 'to tread', we first applied Wechsler's test on the verbal stem *lopen* 'to walk' (16) and then to the entire ICV *belopen* (17).

- (16) Ze liepen, #en als resultaat, raakten ze gelopen.
they walked and as result became they walked
 'They walked, #and as a result they became walked.'
- (17) Ze beliepen de paden, en als resultaat raakten de paden belopen.
they walked the paths and as result became the paths walked
 'They trod paths, and as a result the paths became trodden.'

Given that the verbal stem is not resultative, but the prefix-ICV is, this prefix was annotated as adding resultativity. Note that the non-resultativity of *lopen* in (16) cannot solely be due to it not selecting an object DP; resultativity can also apply to the subject. An example of a resultative meaning targeting the subject is given in (18). We therefore also annotated for subject resultativity in the dataset.

- (18) De gemoederen bedaarden, en als resultaat, raakten de gemoederen
the moods calmed.down, and as result, became the moods
bedaald.
calmed.down
 'The moods calmed down, and as a result the moods became calmed down.'

Whenever a verbal stem was determined to license a resultative interpretation based on Wechsler's test, the effect of the prefix/prefixoid on the ICV was annotated as not resultative.³ In the case of a non-verbal base, only the resultativity of the ICV could be tested. In those cases, we relied directly on the outcome of Wechsler's test on the ICV. In sum, for ICVs with a verbal base, we applied Wechsler's test both on the stem and on the ICV itself, and only in cases in which the stem was not resultative but the ICV was, we annotated a resultative effect of the prefix/prefixoid. For ICVs with a nonverbal base, we could only apply Wechsler's test to the ICV itself, and annotated a resultative effect of the prefix/prefixoid if the ICV was resultative.

Last, we discuss the annotation of Property III, i.e., whether the prefix/prefixoid can alter the argument structure of the stem it attaches to. Similar to our analysis of Property II, we could only determine the argument structure of the stem when this stem was verbal. In all other cases (i.e., nominal stems, adjectival stems, and NLSs), we treated the prefix/prefixoid as altering argument structure, given that its attachment yields a verbal predicate. For ICVs with verbal cases, we annotated for the argument structure of the stem and that of the ICV, and furthermore the auxiliary selected by the stem and the ICV. If one (or both) of these factors showed a change after the attachment of the prefix/prefixoid, we annotated for that particular ICV that the prefix/prefixoid has an

³ A JCGL-reviewer comments that he/she is convinced that in ICVs in which the verbal stem is resultative on its own, it is still the prefix that is syntactically responsible for the resultativity of the ICV. We agree with the reviewer on this theoretical claim, but for the empirical part of this paper we think it is better to annotate for the resultative effect of prefixoids/prefixes rather conservatively without prior theoretical choices influencing the choices made in the annotation.

Table 1 Number of prefixoid-ICVs per prefixoid/category

Prefixoid	Category	Unique ICVs (#)	Total (%)
<i>aan</i> ‘at’	P	3	1.5
<i>achter</i> ‘behind’	P	2	1.0
<i>door</i> ‘through’	P	30	15.4
<i>om</i> ‘around’	P	36	18.5
<i>onder</i> ‘under’	P	31	15.9
<i>over</i> ‘over’	P	58	29.7
<i>voor</i> ‘in front of’	P	5	2.6
<i>mis</i> ‘wrong’	A	16	8.2
<i>vol</i> ‘full’	A	6	3.1
<i>weer</i> ‘again’	Adv	8	4.1
Total		195	100

effect on the argument structure. For example, the verbal stem *kletsen* ‘to chat’ selects a subject but no object DP, as illustrated in (19). The prefix-ICV *be-kletsen* ‘to chat about something’ does select an object DP (20). This means that for the prefix-ICV *be-kletsen* we annotated that the prefix has an effect on argument structure.

(19) Ze kletsten (*de man).
they chatted the man
 ‘They chatted.’

(20) Ze bekletsten de man.
they PREF-chatted the man
 ‘They chatted about the man.’

When the dictionary provided multiple entries for a verbal stem/ICV, the first entry was used, or, in the case of the verbal stem, the entry whose semantics is closest to the semantics of the ICV. With the annotation techniques in place, we now move on to the results, which we discuss below.

2.2 Results

As presented above, both the sample of prefixoid-ICVs and of prefix-ICVs were annotated for three properties (Properties I–III). In what follows, we present the results of each of these properties, after first discussing the single property which was only annotated for in the prefixoid-ICV sample, namely the category of the prefixoid. We end this subsection with a summary of the most important results.

2.2.1 Category of the prefixoid

The number of unique prefixoid-ICVs that were found in the corpus per prefixoid are given in Table 1.

Table 2 Category of stem in prefixoid-ICVs

Prefixoid	V <i>n</i> (%)	N <i>n</i> (%)	A <i>n</i> (%)	NLS <i>n</i> (%)
<i>aan</i> ‘at’	2 (66.7)	0 (0)	0 (0)	1 (33.3)
<i>achter</i> ‘behind’	2 (100)	0 (0)	0 (0)	0 (0)
<i>door</i> ‘through’	28 (93.3)	2 (6.7)	0 (0)	0 (0)
<i>om</i> ‘around’	21 (58.4)	13 (36.1)	0 (0)	2 (5.5)
<i>onder</i> ‘under’	29 (93.5)	2 (6.5)	0 (0)	0 (0)
<i>over</i> ‘over’	46 (79.3)	8 (13.8)	0 (0)	4 (6.9)
<i>voor</i> ‘in front of’	5 (100)	0 (0)	0 (0)	0 (0)
<i>mis</i> ‘wrong’	16 (100)	0 (0)	0 (0)	0 (0)
<i>vol</i> ‘full’	6 (100)	0 (0)	0 (0)	0 (0)
<i>weer</i> ‘again’	8 (100)	0 (0)	0 (0)	0 (0)
Total (%)	79.0	17.4	0	3.6

The number of unique occurrences is highest among the P-based prefixoids, even though quite a few P-based prefixoids have very low numbers of unique occurrences as well. With these data in place, we move on to the results of the annotation of the three properties under discussion, starting with the category of the stem to which the prefixoids/prefixes attach.

2.2.2 Category of the stem

Let us first look at the categories of the stems to which prefixoids attach. The data are presented in Table 2.

The majority of the stems in prefixoid-ICVs are verbal; this holds for 79% of the data set. Nominal stems are much less frequent at 17.4% of the data set. Adjectival stems were not found. Finally, in very few cases, the prefixoid attaches to a NLS, 3.6% of the data set. Two examples of prefixoid-ICVs with a verbal stem are given in (21)–(22), two examples with a nominal stem in (23)–(24), and two examples with a NLS in (25)–(26).

- (21) a. zoeken
search
‘to search’
b. door-zoek-en
through-search-INF
‘to search through’
- (22) a. galmen
echo
‘to echo’
b. weer-galm-en
again-echo-INF
‘to reverberate’

Table 3 Category of stem in prefix-ICVs

Prefix	V <i>n (%)</i>	N <i>n (%)</i>	A <i>n (%)</i>	NLS <i>n</i>
<i>be-</i>	109 (72.6)	22 (14.7)	7 (4.7)	12 (8.0)
<i>ver-</i>	82 (54.7)	17 (11.3)	34 (22.6)	17 (11.4)
<i>ont-</i>	73 (73.0)	19 (19.0)	3 (3.0)	5 (5.0)
Total (%)	65.8	14.5	11	8.7

- (23) a. arm
 arm
 ‘arm’
 b. om-arm-en
 around-arm-INF
 ‘to embrace’
- (24) a. titel
 title
 ‘title’
 b. onder-titel-en
 under-title-INF
 ‘to provide with subtitles’
- (25) a. vaard
 NLS
 b. aan-vaard-en
 at-NLS-INF
 ‘to accept’
- (26) a. rompel
 NLS
 b. over-rompel-en
 over-NLS-INF
 ‘to overwhelm’

What is important to note from Table 2 is that not all prefixoids show identical behavior with respect to the types of stems they can attach to. That is, all the A-based and Adv-based prefixoids (i.e., *mis* ‘wrong,’ *vol* ‘full,’ and *weer* ‘again’) only attach to verbal stems. The same holds for the P-based prefixoids with very low occurrences in the data set: *voor* ‘in front of’ and *achter* ‘behind’. The four P-based prefixoids which show higher unique occurrences in the data set, namely *door* ‘through,’ *om* ‘around,’ *onder* ‘under,’ and *over* ‘over’ do attach to nominal stems, with relatively high frequency for *om* ‘around.’

Let us now present the results for the same property of the prefix-ICVs. The results are given in Table 3.

The majority of these stems are verbal: 65.8% of the data set. Note, however, that this is a lower percentage than in the prefixoid-ICV data set (79%). In particular, the prefix *ver-* has a lower frequency of attachment to verbal stems; it also attaches quite

frequently to adjectival stems (22.6%), followed by nominal stems (11.3%) and NLSs (11.4%). Two examples of prefix-ICVs with a verbal stem are given in (27)–(28), two examples with a nominal stem in (29)–(30), two examples with an adjectival stem in (31)–(32), and two examples with a NLS in (33)–(34).

- (27) a. *knaag-en*
 nibble-INF
 ‘to nibble’
 b. *be-knaag-en*
 PREF-nibble-INF
 ‘to gnaw’
- (28) a. *zwell-en*
 swell-INF
 ‘to swell’
 b. *ont-zwell-en*
 PREF-swell-INF
 ‘to make less swollen’
- (29) a. *eelt*
 callus
 ‘callus’
 b. *ver-eelt-en*
 PREF-callus-INF
 ‘to get covered by callus’
- (30) a. *bos*
 forest
 ‘forest’
 b. *ont-boss-en*
 PREF-forest-INF
 ‘to deforest’
- (31) a. *nieuw*
 new
 ‘new’
 b. *be-nieuw-en*
 PREF-new-INF
 ‘to be/make curious’
- (32) a. *mooi*
 beautiful
 ‘beautiful’
 b. *ver-mooi-en*
 PREF-beautiful-INF
 ‘to become/make beautiful’
- (33) a. *geet*
 NLS
 b. *ver-geet-en*
 PREF-NLS-INF

Table 4 Resultative effect of prefixoids in prefixoid-ICVs

Prefixoid	Adds resultativity <i>n</i> (%)	Does not <i>n</i> (%)
<i>aan</i> ‘at’	2 (66.7)	1 (33.3)
<i>achter</i> ‘behind’	0 (0)	2 (100)
<i>door</i> ‘through’	13 (43.3)	17 (56.7)
<i>om</i> ‘around’	22 (43.3)	13 (38.9)
<i>onder</i> ‘under’	5 (16.1)	26 (83.9)
<i>over</i> ‘over’	26 (44.8)	32 (55.2)
<i>voor</i> ‘in front of’	0 (0.0)	4 (100.0)
<i>mis</i> ‘wrong’	0 (0.0)	16 (100.0)
<i>weer</i> ‘again’	0 (0.0)	8 (100.0)
<i>vol</i> ‘full’	0 (0.0)	6 (100.0)
Total <i>n</i> (%)	70 (35.9)	125 (64.1)

‘to forget’

- (34) a. edig
NLS
b. be-edig-en
PREF-NLS-INF
‘to swear in’

In sum, with respect to Property I, the data showed the following characteristics: The vast majority of the stems to which prefixoids attach are verbal; this percentage is higher compared to that of the prefix-ICV data set. Prefixoids never attach to adjectival stems, and very rarely to NLSs. Prefixes do attach to adjectival stems, and more frequently to NLSs compared to prefixoids. Among the prefixoids, the A-based and Adv-based prefixoids only attach to verbal stems, whereas some of the P-based prefixoids can also attach to nominal stems and NLSs.

2.2.3 Resultative effect

We now turn to Property II, namely whether and how frequently prefixoids and prefixes have a resultative effect on the compositional semantics of the ICV. The percentages of prefixoid-ICVs in which the prefixoid adds resultativity or does not contribute resultativity are given in Table 4.

Two examples of prefixoid-ICVs in which the prefixoid adds a resultative effect are given in (35)–(36). Two examples in which the prefixoid has no effect on resultativity are provided in (37)–(38).

- (35) a. Ze cirkelden door de lucht, #en als resultaat raakten ze gecirkeld.
they circled through the air and as result became they circled
‘They circled through the air, #and as result, they became circled.’

- b. Ze om-cirkelden de vragen, en als resultaat raakten de
they around-circled the questions and as result became the
 vragen om-cirkeld.
questions around-circled
 ‘They encircled the questions, and as a result, the questions became encircled.’
- (36) a. Ze dachten lang, #en als resultaat raakten ze gedacht.
they thought long and as result became they thought
 ‘They thought for long, #and as a result, they became thought.’
 b. Ze door-dachten het probleem, en als resultaat raakte het
they through-thought the problem, and as result became the
 probleem door-dacht.
problem through-thought
 ‘The thought the problem through, and as a result, the plan became thought through.’
- (37) a. Ze spekten de kas, en als resultaat raakte de kas
they filled the cash.register and as result became the cash.register
 gespekt.
filled
 ‘They filled the cash register, and as a result, the cash register became filled up.’
 b. Ze doorspekten het verhaal met leugens, en als resultaat raakte
they through-filled the story with lies and as result became
 het verhaal door-spekt met leugens.
the story through-filled with lies
 ‘They filled the stories with lies, and as a result, the story became filled up with lies.’
- (38) a. Ze laadden de vrachtwagen, en als resultaat raakte de vrachtwagen
they loaded the truck and as result became the truck
 geladen.
loaded
 ‘They loaded the truck, and as a result, the truck became loaded.’
 b. Ze over-laadden de vrachtwagen, en als resultaat raakte de
they over-loaded the truck and as result became the
 vrachtwagen over-laden.
truck over-loaded
 ‘They overloaded the truck, and as a result, the truck became overloaded.’

As can be seen in Table 4, in the majority of examples in the data set, the prefixoid does not add resultativity to the compositional semantics of the ICV (64.1%). However, it is also not the case that prefixoids never contribute resultativity. Here we see, as was the case for attaching solely to verbal stems (Property I), that the A-based and Adv-based prefixoids behave uniformly in that they never have a resultative effect. The P-based prefixoids can, on occasion, have a resultative effect. In particular, *onder* ‘under’, *om* ‘around’, and *over* ‘over’ often add resultativity. Before we move on, it is important

Table 5 Overall resultativity of prefixoid-ICVs

	Resultative ICVs	Non-resultative ICVs
<i>n</i>	180	15
%	92.3	7.7

Table 6 Resultativity of prefixes in prefix-ICVs

Prefix	Adds resultativity <i>n</i> (%)	Does not <i>n</i> (%)
<i>be</i>	85 (56.7)	65 (43.3)
<i>ver</i>	102 (68.0)	48 (32.0)
<i>ont</i>	41 (41.0)	59 (59.0)
Total <i>n</i> (%)	228 (57.0)	172 (43.0)

to mention that the overall frequency of resultative semantics of prefixoid-ICVs is very high; only 15 tokens in the data set (7.7%) are prefixoid-ICVs without resultative semantics. The relevant numbers are shown in Table 5.

There is thus a strong preference for prefixoids to attach to stems that are already resultative. This is true despite the fact that they either never (A-/Adv-based prefixoids) or only sometimes (P-based prefixoids) add resultativity through their co-occurrence with particular stems.

Moving on to the resultative effect of prefixes, we provide results with respect to whether the prefix in the prefix-ICV adds or does not add resultativity in Table 6.

Two examples of prefix-ICVs in which the prefix adds resultativity are provided in (39)–(40). Two examples of prefix-ICVs where the prefix has no additional resultativity effect are given in (41)–(42).

- (39) a. Het donderde, #en als resultaat raakte het gedonderd.
it thundered and as result became it thundered
 ‘It thundered, and as a result, #it became thundered.’
 b. Ze be-donderden ons, en als resultaat raakten we be-donderd.
they PREF-thundered us and as result became we REF-thundered
 ‘They fooled us, and as a result, we got fooled.’
- (40) a. De kinderen rolden door de tuin, #en als resultaat raakten ze
the children rolled through the garden and as result became they
gerold.
rolled
 ‘The children rolled through the garden, #and as a result, they became rolled.’
 b. Ze ont-rolden de loper, en als resultaat raakte de loper
they PREF-rolled the carpet and as result became the carpet
ont-rolld.
PREF-rolled
 ‘They unrolled the carpet, and as a result, the carpet got unrolled.’

Table 7 Overall resultativity of prefix-ICVs

	Resultative ICVs	Not resultative ICVs
<i>n</i>	385	15
%	92.2	3.8

- (41) a. Ze filmde de situatie, en als resultaat raakte de situatie gefilmd.
they filmed the situation and as result became the situation filmed
 ‘They filmed the situation, and as a result, the situation got filmed.’
 b. Ze ver-filmde het boek, en als resultaat raakte het boek ver-filmd.
they PREF-filmed the book and as result became the book filmed
 ‘The made a movie out of the book, and as a result, the book got turned into a movie.’
- (42) a. Ze zoutten het vlees, en als resultaat raakte het vlees gezout.
they salted the meat and as result became the meat salted
 ‘They salted the meat, and as a result, the meat became salted.’
 b. Ze ont-zoutten het zeewater, en als resultaat raakte het zeewater
they PREF-salted the sea.water and as result became the sea.water
 ont-zout.
PREF-salted
 ‘They desalted the sea water, and as a result, the sea water became desalted.’

As shown by the examples in Table 6, in the majority of the data set (57.0%), the prefix adds the resultative meaning component to the compositional semantics of the ICV. Out of the three prefixes, *ver-* does so the most (in 68.0% of *ver*-ICVs) and *ont-* the least (in 41.0% of *ont*-ICVs). Moreover, as was the case in the prefixoid-ICV data set, it is important to mention that in the entire data set of prefix-ICVs there are only 15 ICVs that do not have a resultative meaning (3.8% of the data set). The relevant numbers are shown in Table 7.

This means that prefixes generally pair with ICVs that are already resultative, irrespective of whether or not they add the resultativity themselves or whether it is part of the semantics of the stem they attach to.

Summarizing our findings regarding Property II, both prefixoids and prefixes show a strong preference for combining with a resultative ICV. Prefixoids add this resultativity themselves less frequently than do prefixes.

Among the group of prefixoids, A-based and Adv-based prefixoids never add a resultative meaning. Among the group of prefixes, *ver-* most frequently adds the resultative meaning itself, whereas *ont-* does so the least.

2.2.4 Argument structure

Finally, we present the results regarding Property III, namely whether and how frequently prefixoids/prefixes have an effect on the argument structure of the stem. We start with the results regarding the effect on argument structure of prefixoids in prefixoid-ICVs. These are presented in Table 8.

Table 8 Effect of prefixoids on argument structure

Prefixoid	Affects argument structure <i>n</i> (%)	Does not <i>n</i> (%)
<i>aan</i> ‘at’	3 (100)	0 (0)
<i>achter</i> ‘behind’	0 (0)	2 (100)
<i>door</i> ‘through’	13 (43.3)	17 (56.7)
<i>om</i> ‘around’	27 (75.0)	9 (25.0)
<i>onder</i> ‘under’	8 (25.8)	23 (74.2)
<i>over</i> ‘over’	32 (55.2)	26 (44.8)
<i>voor</i> ‘in front of’	1 (20.0)	4 (80.0)
<i>mis</i> ‘wrong’	8 (50.0)	8 (50.0)
<i>vol</i> ‘full’	2 (33.3)	4 (66.7)
<i>weer</i> ‘again’	2 (25.0)	6 (75.0)
Total <i>n</i> (%)	96 (49.2)	94 (50.8)

Two examples of prefixoid-ICVs in which the prefixoid changes the argument structure of the stem are given in (43)–(44). (43) shows a case in which an object DP is added, and (44) shows a case in which the object DP becomes obligatorily reflexive. Two examples of prefixoid-ICVs in which the prefixoid does not change the argument structure of the stem are provided in (45)–(46). In both cases, an object DP is selected.

- (43) a. *Ze bidden* (*de man).
they pray the man
‘They pray.’
b. *Ze aan-bidden* de man.
they at-pray the man
‘They worship the man.’
- (44) a. *Ze dragen* de tassen.
they carry the bags
‘They carry the bags.’
b. *Ze mis-dragen* zich/*de tassen.
they wrong-carry REFL/the bags
‘They misbehave themselves.’
- (45) a. *Ze kruisen* de lijnen.
they cross the lines
‘They cross the lines.’
b. *Ze door-kruisen* de woestijn.
they through-cross the desert
‘They cross the desert.’
- (46) a. *Ze houden* de hond.
they keep the dog
‘They keep the dog.’

Table 9 Effect of prefixes on argument structure

Prefix	Affects argument structure <i>n</i> (%)	Does not <i>n</i> (%)
<i>be</i>	108 (72.0)	42 (28.0)
<i>ver</i>	107 (71.3)	43 (28.7)
<i>ont</i>	59 (59.0)	41 (41.0)
Total <i>n</i> (%)	274 (68.5)	126 (31.5)

- b. Ze onder-houden het huis.
they under-keep the house
 ‘They maintain the house.’

According to Table 8, in about half of the data set, the prefixoid has an effect on the argument structure of the stem (49.2%). For this property, we do not observe any striking differences between the A-based and Adv-based prefixoids versus the P-based prefixoids. It is worth highlighting that the A-based prefixoid *mis* ‘wrong’ often has an effect on the argument structure of the stem, in roughly half of the cases. Note, however, that this effect always changes an object DP into a reflexive object DP (as illustrated in (46)), which is arguably a less striking effect than actually adding or removing an argument from the argument structure.

Let us now investigate the effect of the prefixes on argument structure in prefix-ICVs. The results are presented in Table 9.

Two examples of prefix-ICVs in which the prefix changes the argument structure of the stem are given in (47)–(48). Both sets of examples include cases in which an object DP is added to the argument structure of the stem. Two examples of prefix-ICVs in which the prefix does not have an effect on the argument structure of the stem are illustrated in (49)–(50).

- (47) a. Ze gluren urenlang (*de man).
they peek hours.long the man
 ‘They are peeking for hours.’
 b. Ze be-gluren de man.
they PREF-peek the man
 ‘They are peeking at the man.’
- (48) a. Ze zeilen urenlang (*de problemen).
they sail hours.long the problems
 ‘The sailed for hours.’
 b. Ze ont-zeilen de problemen.
they PREF-sail the problems
 ‘They are avoiding the problems by sailing away.’
- (49) a. Ze keren het schip.
they turn the ship
 ‘They are turning the ship.’
 b. Ze be-keren de inwoners.
they PREF-turn the inhabitants

‘They are converting the inhabitants.’

- (50) a. Ze zwachtelen het been.
 they bandage the leg
 ‘They bandage the leg.’
 b. Ze ont-zwachtelen het been.
 they PREF-bandage the leg
 ‘They unwrap the leg.’

Based on the data in Table 9, in the majority of instances, the prefix has an effect on the argument structure of the stem it attaches to (68.5% of the dataset). This is a higher percentage than we find in the prefixoid-ICV dataset (49.2%).

Overall, the results regarding Property III are in line with the general findings with respect to the other two properties, namely, that prefixes outperform prefixoids when it comes to all three properties.

2.2.5 Data summary

The results of the annotation of data from the *Celex Lexicon* for Properties I–III revealed the following patterns: First, with respect to Property I (the category of the stem to which prefixoids/prefixes attach), it was shown that prefixoids never attach to adjectival stems, very rarely to NLSs, infrequently to nominal stems, and most frequently to verbal stems. Furthermore, A-based and Adv-based prefixoids only attach to verbal stems. Prefixes can attach to all four types of stems.

Second, with respect to Property II (whether or not prefixoids/prefixes have a resultative effect on the compositional semantics of the ICV they are part of), it was shown that prefixoids add resultativity less frequently than prefixes. A-based and Adv-based prefixoids never add resultativity. The (absence of a) resultative effect of the prefixoid/prefix notwithstanding, the majority of ICVs in both datasets are resultative. This means that in those cases in which a prefixoid/prefix does not add or contribute to the resultativity of the event, it still strongly prefers to attach to stems with inherent resultative semantics.

Third, concerning Property III (whether or not prefixoids/prefixes have an effect on the argument structure of the stem they attach to), it was shown that overall, prefixoids have an effect on the argument structure of the base less often than prefixes do.

In summary, we have seen that with regards to the three investigated properties, prefixes outperform prefixoids in all three respects. Based on these observations, it seems to be the case that prefixes have a strong syntactic and semantic effect within the first phase, whereas most prefixoids do not. Within the group of prefixoids, A-based and Adv-based prefixoids have virtually no effect on the syntax and semantics of the first phase. They never attach to nonverbal stems, which indicates that they cannot verbalize, nor are they the cause of the resultative semantics of the ICV that they are part of. Although they do sometimes alter the argument structure of the stem they attach to, this mostly involves an object DP being selected as an obligatory reflexive (object) DP.

3 Theoretical analysis

In this section, we present our theoretical analysis of prefixoid- and prefix-ICVs in Dutch. In Sect. 3.1, we lay out our theoretical prerequisites, namely (i) the event decomposition as proposed by Ramchand (2018) and (ii) some key components of the analysis of Folli and Harley's 2020 account of verb framed and satellite framed languages, on which we will build our analysis. In Sect. 3.2, we propose that the crucial distinction between Dutch prefixoids and prefixes in ICVs can be accounted for based on three structural differences in their internal and external syntax. We argue therefore that (i) prefixoids are clitics, whereas prefixes are not, (ii) clitics are adjoined via Pair Merge, whereas prefixes enter the structure via Set Merge, and (iii) prefixoids are attached high (at the edge of ν P), whereas prefixes are attached low (lexicalizing Res, to which a root is adjoined prior to verbalization).

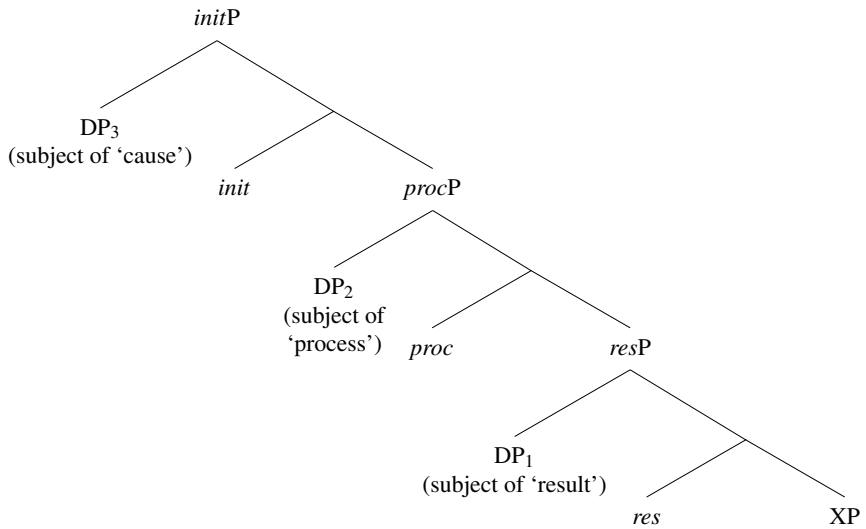
Equipped with the key parts of Folli and Harley (2020) and the structural distinction between prefixoids and prefixes, we discuss our analysis of prefixoid- and prefix-ICVs in Dutch in Sect. 3.3.

3.1 Theoretical prerequisites

3.1.1 Lexicalizing resultativity

In our analysis, we adopt a fine-grained decomposed event structure of the first phase (ν P) as developed and advocated by Ramchand (2008, 2019) among others. According to the tree structure in (51), the event semantics associated with the first phase consist of functional heads that establish (i) the initiator of an event (*init*P), (ii) the process (or dynamic nature) of an event (*proc*P), and (iii) the result state of an event (*res*P). We adopt the functional projection ResP in our analysis of Dutch prefix-ICVs.

(51) Decomposed event structure of the first phase (vP, Ramchand 2008)



3.1.2 Folli and Harley (2020)

Folli and Harley (2020) present a head movement approach to Talmy's typology of verb-framed (VF)/satellite-framed (SF) languages (Talmy 1975, 1985).⁴ Dutch, like English, is a SF language (Hoekstra et al. 1987), in which Result can be expressed both by the verb and by a satellite, unlike VF languages (e.g., Italian) in which Result can only be lexicalized by the verb. To illustrate that Dutch is a SF language, consider the examples below: in (52) the result state is lexicalized by the adjective, and in (53) by the particle of a particle verb. In VF languages, such structures are impossible.⁵

⁴ Folli and Harley's (2020) account of VF/SF variation has recently been challenged by Acedo-Matellán and Kwapiszewski (2024), in which they put forward an alternative account which involves spanning. A JCGL-reviewer wonders why we don't build our analysis on this more recent account, in which Dutch ICVs like *be-planten* lit. *prf-plant* are briefly discussed and analysed as well. The answer is twofold. First, in their analysis of such verbs in Dutch, Acedo-Matellán and Kwapiszewski (2024) analyse the prefix as a root rather than directly lexicalizing Res. We don't want to assume Dutch prefixes contain a root, however, due to their clearly functional semantics. Second, in their analysis, the prefix only attaches after the root has been verbalized, a structural choice which we don't want to adopt due to the fact that such structures predict that all roots which occur in prefix-ICVs can also serve as standalone verbs, contrary to fact (see among others Wood (2023) and Šereikaitė (2017) for discussion of different attachment heights of prefixes within the first phase). That being said, we do think the analysis of Acedo-Matellán and Kwapiszewski (2024) overall is very promising, and perhaps in future work their analysis can be fruitfully extended to prefix-ICVs, but as the analysis stands now, we are not ready to adopt their suggested analysis for Dutch prefix-ICVs.

⁵ A rich body of generative analyses exist to account for the differences between VF and SF languages, which also extend to other parts of Talmy's typology we don't address here, namely the lexicalization of Manner and Motion. We refer the interested reader to Folli and Harley (2020) for a detailed overview of several types (both in terms of empirical scope and theoretical perspective) of previous analyses.

- (52) De man schildert de muur **rood**.
the man paints the wall red
 ‘The man paints the wall red.’
- (53) De man drinkt de melk **op**.
the man drinks the milk up
 ‘The man finishes the milk.’

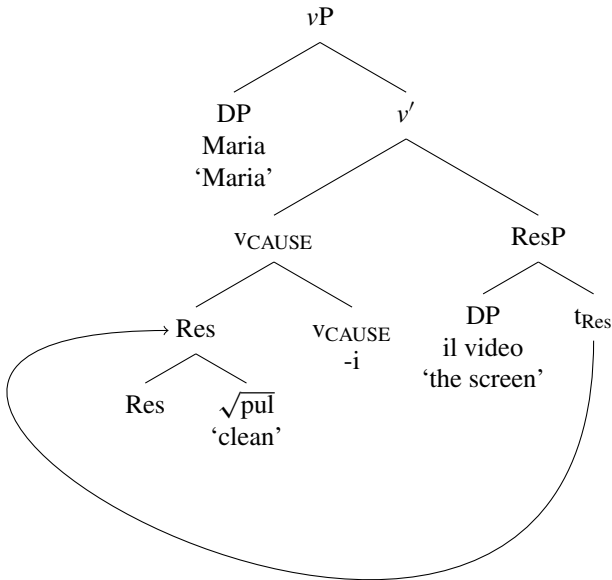
Folli and Harley (2020, 444) propose that the fact that VF languages are not able to lexicalize Result via elements other than the verb is due to a specific type of head movement parameter, namely a ‘Result-to-*v*’ parameter. More specifically, they argue that in VF languages, certain flavors of *v* (along with change-of-state *v* that select for a Result) come with an uninterpretable Res feature. This feature is specified with a movement-triggering EPP feature that forces Res to move to *v*. This means that Res cannot be lexicalized by an adjective or particle, for example. In other words, in a language that has this parameter switched on, Result can only be expressed by the verb. For SF languages, Folli and Harley (2020) argue that the lexicon contains both a change-of-state *v* with an EPP-bearing [*u*Res] as well as one without the EPP-feature. Put differently, Res can be moved to *v* in some cases (e.g., Result being expressed by the verb), but in other cases Res will be stranded and lexicalized independently from the verb.

In our analysis, we directly build on their proposal, and extend this to Dutch prefix-ICVs. For our purposes, a specific footnote in Folli and Harley (2020, 457) is of particular relevance, where they state:

Note that in previous analyses of bimorphemic change-of-state verbs involving particles such as *di-* or *in-*, it has been proposed that the particle realizes a Result while the root $\sqrt{\text{ }}$ is a *v*-adjoined manner element, with incorporation of Res into *v* creating the complex verb $[[\text{de}]_{\text{Res}}-[\text{stroy}] \emptyset_v]_v$ (Marantz 2001; Harley 2008). The proposal made here entails that this is not correct. Instead, **these predicates must involve adjunction of the root element to the Res head realized by the particle; they are thus categorized by Res, not by *v*. Subsequent head movement of Res to *v* recategorizes the whole Res complex, as in (38).** (Boldface ours)

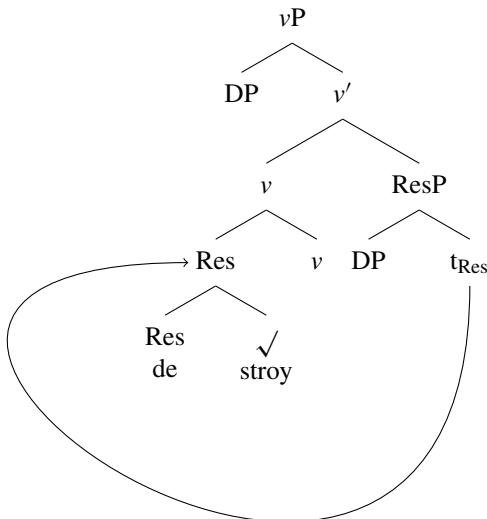
For clarity, their structure in (38) for the Italian *vP* *Maria he pulito il video* (Eng. ‘Maria cleaned the screen’) is reproduced below in (54) (Folli and Harley 2020, 448).

(54) vP structure of ‘Maria ha pulito il video’



Translating this structure into the structure they have in mind for bimorphemic change-of-state verbs like ‘destroy’ in English gives use the following structure:

(55) Incorporation of Res-to- v in bimorphemic verbs in English



The analysis suggested by Folli and Harley (2020) has direct consequences for our analysis of Dutch ICVs. Building on their proposal, we propose that the prefix in Dutch ICVs is capable of lexicalizing the Res-head, i.e., the prefix in prefix-ICVs is

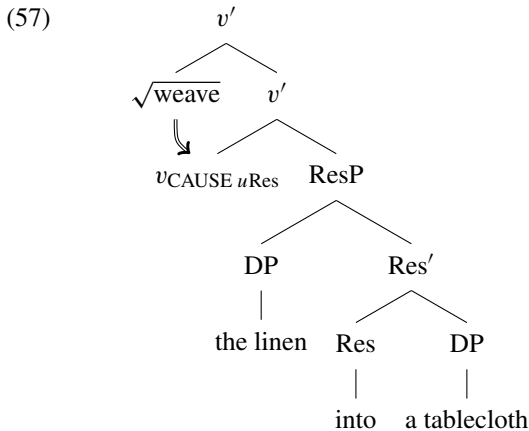
the Res-head in (55), and is directly Merged with the root, prior to verbalization. In contrast, as we will argue in the next section, prefixoids can never be realized as the head of ResP and, due to their status as clitics, must head-adjoin with the phase head v .⁶

However, before moving on to the specifics of our own analysis, we need to make clear that there is one particular part of Folli and Harley's (2020) account of bimorphemic English verbs that we do not adopt in our own analysis of Dutch prefix-ICVs, namely their suggestion that such bimorphemic verbs require the assumption of a morphological filter. Recall that in their account, English change-of-state v is underspecified for a movement triggering EPP-feature, which they translate into the lexicon having both a change-of-state v specified for $[uRes]^*$ and one specified for $[uRes]$ (i.e., without EPP-feature). With respect to bimorphemic verbs like 'destroy', they argue that this underspecification becomes problematic. How can we avoid Res (e.g., when lexicalized by 'de-') being selected by the change-of-state v specified for $[uRes]$ that does not bear an EPP-feature? Such a combination would result in the stranding of prefix 'de-', which clearly should be ruled out. They propose that such a configuration is in fact possible in the syntax of English, but that at the point of Vocabulary Insertion, the root of a bimorphemic verb (e.g., $\sqrt{pose}$) is conditioned by one or more particular particles in Res. For example, for the root $\sqrt{pose}$ they propose it is a contextually conditioned Vocabulary item, its insertion conditions given in (56):

$$(56) \quad \sqrt{\quad} \Leftrightarrow [[\{com-, im-, re-, de-, trans-, op-, pro-\}_{Res} [_\ v]]]_v$$

We do not adopt this solution for bimorphemic verbs in Dutch, due to the fact that there are simply too many prefix-ICVs in Dutch for it to be captured with such a morphological filter. Instead, we want to argue that the obligatory movement of the Res-root complex (i.e., prefix and root) to change-of-state v in Dutch ICVs is due to a need of this v to categorize syntactic structure. As also mentioned in the quote of Folli and Harley (2020) above, note that in previous DM-analyses of bimorphemic verbs in English, the root was adjoined to v . In the analysis of Folli and Harley (2020), such root-adjunction to change-of-state v is formalized by means of e-Merge followed by m-merger à la Matushansky (2006). As shown above, Folli and Harley (2020) no longer assume this to be the correct analysis for bimorphemic verbs, but do assume e-Merge of Manner roots in other configurations, for example for manner-denoting elements such as $\sqrt{weave}$ or $\sqrt{float}$ in English. Their tree structure for such verbs is given in (57), in which m-merger is indicated by means of a double-lined arrow.

⁶ An anonymous JCGL-reviewer posed the question as to whether or not our analysis more strongly relates to *phrasal layering-* or *complex head-analysis* (Paparounas 2023a, b; Embick 2023; Wood 2023). Based on our discussion in this section, we can say that our analysis of Dutch prefix-ICVs aligns more naturally with a *complex head-analysis*.



In such structures, then, change-of-state v categorizes the root. In the structure for bimorphemic verbs suggested by Folli and Harley (2020) and given above in (55), the root is categorized by Res; no root e-merges to change-of-state v . Our proposal then for the question raised by Folli and Harley (2020) as to how to make sure that the Res-root complex in bimorphemic verbs moves to v is the following: v has to categorize a root-containing structure in syntax, which it can only do locally, i.e., by only either m-merger or by attracting a lower root-containing structure to head-adjoin to itself. In SF languages, this means that Res can be stranded as long as a Manner root is e-merged to change-of-state v , which is in line with Folli and Harley's (2020) analysis for structures such as the one in (57). When this is not the case, namely in bimorphemic verbs/ICVs, change-of-state v will attract a lower root-containing structure in order to verbalize this: the Res-root complex. A detailed implementation of this suggestion is beyond the scope of the current paper, but assuming for now that it is on the right track, we can extend Folli and Harley's (2020) account to Dutch prefix-ICVs without the need for a morphological filter.

With the key components of Folli and Harley (2020) in place, we can now move on to our proposal regarding the internal and external syntax of Dutch prefixoids and prefixes.

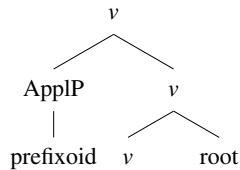
3.2 Structural differences between Dutch prefixoids and prefixes

In our account, prefixoids and prefixes essentially differ from each other with respect to (i) their internal syntax (i.e., whether they are X^0 or XP-elements), (ii) their point of initial merger (i.e., vP -internal or at the edge of the 'first phase' (v /Voice)), and (iii) what sort of Merge (i.e., Set vs. Pair Merge) is involved in introducing them into narrow syntax. We claim here that prefixoids are clitics, adjoined via Pair Merge at the edge of the first phase (Svenonius 2004; Epstein et al. 2016; Šereikaitė 2017; Manzini and Pescarini 2022), whereas prefixes are not clitics, and are Merged vP -internally, lexicalizing Res.⁷

⁷ We thank an anonymous JCGI-reviewer for clarifying this distinction and for helpful references and discussion.

Let us elaborate on the specifics of these distinctions. Turning first to our treatment of prefixoids as clitics, we propose that these elements are min-max categories, since, in principle, they are particles and there is nothing that would ban their existence according to Bare Phrase Structure principles (Chomsky 1995; Lohndal 2014).⁸ We illustrate this process of cliticization by adjunction of the prefixoid with the tree structure in (58):

(58) **Adjunction of prefixoids to v**



In (58) we adopt the label of prefixoids as Appl(icative)P. Admittedly, this label is mostly chosen to reflect the modifier/adjunct status of the prefixoid in prefixoid-ICVs. The key distinction that we want to make here is that P- and/or A/Adv-based elements do not label the vP .⁹

An important debate in the recent literature on clitics concerns the exact nature of their derivation, i.e., whether clitics are derived via Set Merge (which treats the ‘head-like’ status of clitics as fundamentally derived by operations in the phrasal syntax) or whether they are derived via Pair Merge (as items merged directly with phase heads) (Epstein et al. 2016; Manzini and Pescarini 2022). Utilizing Pair Merge to derive clitics, i.e., prefixoids, accounts for their adjunct-like behavior and their label-less status (Wood and Marantz 2017). We therefore propose that Dutch prefixoids have clitic-status and are derived via Pair Merge.¹⁰

In contrast, for Dutch prefixes we assume that they are derived via Set Merge. We furthermore propose that they lexicalize the functional head Res, to which the root of the ICV is adjoined. This means that we propose that the prefix categorizes the root as Res at first. Only later will this Res-root complex be recategorized by v , following Folli

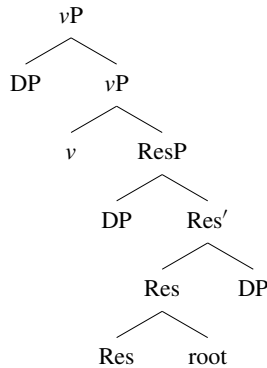
⁸ More concretely, Chomsky (1995, 28) comments that one case in which items can simultaneously be X^0 and an XP is clitics: “In its θ -position, the clitic is an XP; attachment to a head requires it to be an X^0 . The status of clitics as both minimal and maximal projections makes an XP-to-H operation of cliticization possible.”

⁹ The primitive status of adjectives and adverbials in Late-Insertion approaches of the syntax-morphology interface has been the focus of debate in recent studies (Baker 2003; Fábregas 2020; Mitrović and Panagiotidis 2020). Although this interesting debate is (largely) orthogonal to our current study, we have chosen to label prefixoids as Appl(icative)Ps.

¹⁰ We think it is most likely that Pair-Merge only takes place outside of v (rather than inside of its domain). For example, the elements in question are not functional heads, hence, only at something like a phase-edge is where we are likely to expect elements that are not “fully functional.” We thus assume that Pair-Merge is limited to phase heads, which imposes restrictiveness on the system (compared to, say, X-bar theory where adjunction could happen to either head, phrases, or even bar-levels) We thank one of our JCGL-reviewer for kindly pointing this out to us.

and Harley (2020) (see Sect. 3.1.2).¹¹ The merge position for prefixes is illustrated in (59), with the root head-adjoined to it.

(59) **Initial Merge of prefixes (Res)**



To recap, although both prefixoids and prefixes are bound morphemes in ICVs, the former are clitics, and have a much looser connection with the predicates they modify than prefixes, to which the root is directly adjoined. This structural distinction contributes to our explanation of the stacking data presented in the introduction of this paper in examples (6)–(8), which will be discussed in more detail in the next section, together with the discussion of the derivations of Dutch prefixoid- and prefix-ICVs.

3.3 Deriving Dutch prefixoid- and prefix-ICVs

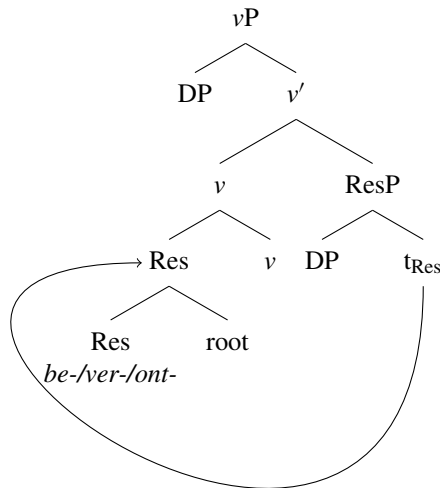
In this section we apply the components of our analysis as laid out in the previous two sections to our Dutch data provided in Sect. 2. First, we discuss the analysis of prefix-ICVs and how the empirical generalizations from Sect. 2 follow from the proposed structure in Sect. 3.3.1. Second, we do the same for prefixoid-ICVs in Sect. 3.3.2, followed by our analysis of the stacking patterns—as discussed at the beginning of this article—in Sect. 3.3.3.

3.3.1 Prefix-ICVs

We claim that all three Dutch prefixes, *be-*, *ver-*, and *ont-*, can function as Res-heads. All prefix-ICVs therefore have the structure given in (60), in which the root has adjoined to Res, and this complex has been head-moved and adjoined to *v*.

(60) **Structure of Dutch prefix-ICV after Res-to-*v* movement**

¹¹ A reviewer points out that the phonological properties of prefixes has led to proposals in the literature that support different structural Merge positions of prefixes (Fenger et al. 2015; Bogomolets 2020; Kouneli et al. 2024). We believe that our analysis of Dutch prefix-ICVs is in line with these recent findings that assume that there is a height difference in prefixes and the relationship with phonological interactions (i.e., contra Julien (2002), who claims prefixes are generally high).



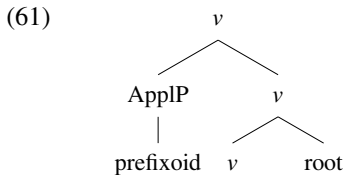
The fact that the three prefixes lexicalize Res explains that they contribute a resultative component to the compositional semantics of the ICV they are part of. Recall, however, that the prefixes do not always do so; they also combine with roots whose lexical semantics is resultative, which means that the prefixes are in principle not needed to render the ICV resultative. We believe this is not a problem for our analysis, though, because the resultative semantics of roots is not syntactic resultativity, which means that in narrow syntax there would not be a double instance of Res.¹² An overlap between lexical features and syntactic features will not cause a crash of the derivation, and will be canceled out at LF.

Let us now discuss the other two properties which we investigated in Sect. 2. Recall that the three prefixes can combine with any type of stem, namely verbal, nominal, adjectival, and NLS. Our analysis can account for these data because in our structure, any root can merge with Res. After this, the Res-complex head-moves and is verbalized by v through incorporation. Recall furthermore our empirical finding that the prefixes often have an effect on the argument structure of the stem they attach to. What we mean with ‘the argument structure of the stem’ here is the idea that the conceptual content of roots are compatible/incompatible with certain structured meanings. These empirical findings are also in line with our analysis. That is, the prefixes lexicalize Res, which always select a DP argument. This means that in cases in which the root’s conceptual content is incompatible with the presence of certain argument, the use of the prefix will make it possible for the ICV to have this additional argument. With this in place, we now move on to the discussion of prefixoid-ICVs and how our analysis fares in explaining the empirical findings presented in Sect. 2.

¹² That is, we assume that some roots have lexical semantics that are compatible with a resultative interpretation, and some do not. The ones that do not, will always need a prefix to receive a resultative interpretation. See Zeijlstra (2008) on the idea that some lexical semantic features overlap in meaning with the meaning of certain functional, syntactic features.

3.3.2 Prefixoid-ICVs

Recall from Sect. 3.2 that we propose prefixoids in prefixoid-ICVs to be clitics, which lexicalize an Appl-head. The structure we propose for prefixoid-ICVs is repeated here in (61).



Due to their status as clitics, they cannot extend beyond ApplP. Let us start with the discussion of the A(dv)-based prefixoids. Recall from Sect. 2 that A(dv)-based prefixoids never add resultativity, never combine with a non-verbal stem, and almost never have an effect on the argument structure. We suggest that these are lexical elements that (i) contain roots and (ii) are categorized as ApplPs.¹³ These roots incorporate into Appl-heads, but adjoin with *v* (i.e., a phase head), which blocks them from starting their own internal functional sequence. Due to their status as a clitic version of these lexical items, these elements are more reduced—or ‘more minimal’—than their more elaborated adjectival or adverbial counterparts that appear in other syntactic contexts. This is, for example, clear from the stress pattern in ICVs; the prefixoid never bears stress, which often results in vowel reduction, compared to the full vowels in their lexical uses. Given that they are adjoined as clitics to *v*, it is expected that they only combine with roots that have already been verbalized, and that they do not alter the argument structure of that verbalized structure.¹⁴ Given that they do not lexicalize a Res head, it follows that they do not add resultative meaning to the compositional semantics of the ICV.

Turning now to the P-based prefixoids, we suggest that those that can have a resultative effect, namely *aan* ‘at,’ *door* ‘through,’ *om* ‘around,’ *over* ‘above’ and *onder* ‘under’ are memorized as indivisible roots, i.e., that univertation between the root and prefixoid has taken place. We think this is a plausible analysis, given the low number of P-based prefixoids that add a resultative interpretation. We propose the same solution for the low number of cases in which a P-based prefixoid combines with a nonverbal stem or alters the argument structure; these numbers are fairly low and are thus very likely to be stored as indivisible units rather than as productive derivational patterns.

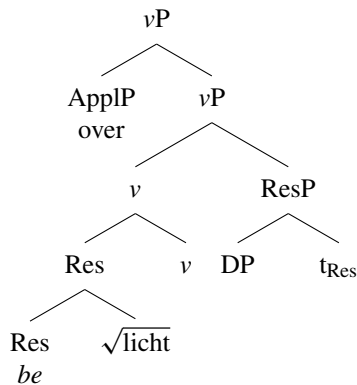
¹³ A reviewer wonders how contentful elements such as Dutch *vol* (i.e., one of the prefixoids in Dutch prefixoid-ICVs) can ever lexicalize a functional head like Appl. We tentatively propose that prefixoids might be semi-lexically used elements, and suggest an analysis along the lines of the approach to semi-lexicity of Cavarani-Pots (2020), in which semi-lexically used elements are roots which are combined with a functional head in a separate workspace, and then inserted into the current workspace as a complex head.

¹⁴ If there appear to be cases where attachment of a prefixoid affects argument structure, our approach predicts that such cases must involve additional structural mechanisms. We refer the reader to Benz (2025) discussion of some related patterns and possible complexities in such cases.

3.3.3 Prefixoid-prefix stacking

Finally, let us return to the stacking data that were presented at the beginning of the paper, which showed that only prefixoid-prefixing is allowed in Dutch ICVs. In our analysis, we have proposed that Dutch prefixes lexicalize a Res head. We adopted a structural analysis proposed by Folli and Harley (2020) for bimorphemic verbs in English (like *destroy*) which we extended to Dutch prefix-ICVs. Recall that in the structure for prefix-ICVs, the prefix is the Res head, to which the root adjoins. This complex then head-moves to *v*, which verbalizes the complex. Given our proposed analysis of prefixoids as clitics which adjoin to *v*, it follows that prefixoid-prefix-stacking is possible. For example, for the ICV with stacking, *overbelichten* ‘overexpose’, the structure would be the following (62).

(62) **Prefixoid-prefix stacking in Dutch**



In this structure, the prefix *be-* lexicalizes Res, to which the root of the ICV adjoins. The complex then moves to *v*, which then labels the entire complex as verbal. This structure is spelled-out, since the first phase is now finished. Building on Uriagareka’s (1999) cyclic spell-out domains, we claim that the prefixoid in this structure is only adjoined (via Pair Merge) after the first cycle of spell-out. This is the only configuration in which two pre-elements can co-exist within the first phase. Res can only be lexicalized once; hence, prefix-prefix stacking is ruled out. The only aspect that the current analysis does not directly account for is the ban on prefixoid-prefixoid stacking, given that the type of adjunction we propose for prefixoids could be applied recursively. We leave this puzzle for future research; however, we want to tentatively suggest that in Dutch, *v* has only one dedicated edge position for modifier-like elements which are not XP-elements. Additional evidence in support of this claim can be found in recursive modification in the nominal domain in Dutch, which is quite common (de Haas and Trommelen 1993; Smessaert et al. 2019), but not so in the verbal domain (Ackema 1999). For example, verbal compounds can only consist of one modifying element (e.g., *mast-klimm-en* lit. mast climb *inf*) ‘to climb masts’, but not *snel-mast-klimm-en* lit. quickly mast climb *inf*). We leave a full analysis of the ban on prefixoid-prefixoid stacking in Dutch ICVs for future work.

4 Conclusion

The goal of this paper was to advance our understanding of the structural differences that exist between prefixoids and prefixes in Dutch ICVs. In the introduction, we discussed prefixoid-prefix-stacking patterns, an observation which seemed to suggest that prefixoids and prefixes occupy different structural positions. We conducted an in-depth annotation study of 195 prefixoid-ICVs and 400 prefix-ICVs, focusing on three properties which relate to the syntax and semantics of the first phase. The results of our annotation showed that, generally, prefixes have a strong effect on the syntax and semantics of the first phase compared to prefixoids, with respect to attaching to nonverbal stems, adding resultativity and adding an argument. Furthermore, within the set of prefixoids, we found a difference in behavior between A(dv)-based prefixoids and P-based prefixoids. The former never have an effect on the first phase, whereas some of the P-based prefixoids seem to have an effect, e.g., sometimes combining with a nonverbal stem, adding resultativity, or affecting the argument structure of the stem.

Approaching these general differences between prefixes and prefixoids in Dutch ICVs from a decompositional perspective to the syntax-morphology interface, we claimed that the key difference between prefixoids and prefixes can primarily be reduced to the clitic-nature of the former. Prefixes in ICV structures are not clitics, and lexicalize a Res head, to which the root adjoins. This takes place lower in the first phase, i.e., prior to the categorization of the Res-root complex by *v*. With this analysis we built upon the analysis of bimorphemic English verbs as suggested by Folli and Harley (2020) in their account of VF/SF variation. For the cases in which a P-based prefixoid does affect the first phase, we claimed that this is only apparent and not syntactic; we argued that in these (limited) cases, the root and prefixoid are memorized as one unit, and hence the resultative semantics/conceptual content compatible with certain argument structure comes from the root. Given the low occurrences of prefixoid-ICVs with resultative meaning, argument structure change or a nonverbal stem, it is not hard to conceive that their specific meaning components can be stored in an idiomatic way. Our analysis provides a straightforward account of the stacking behavior of these elements as well, even though future work is still needed to explain the ban on prefixoid-prefixoid stacking.

Another avenue for future research would be to expand our investigation to include other West Germanic languages which have ICVs, and in particular German and Afrikaans. Both languages have ICVs with prefixoids and prefixes (Van Huyssteen 2020; Wurmbrand 2000), but have never been investigated with respect to the core properties for which we annotated in our annotation study. Furthermore, particularly with respect to Afrikaans, recent work on comparing derivational morphology in Dutch and Afrikaans has shown that patterns that look similar on the surface, on closer scrutiny reveal crucial distinctions in underlying structure (see, e.g., Cavarani-Pots and De Villiers 2023). P-based prefixoids in ICVs in German and Afrikaans are discussed in the literature cited above, but whether A(dv)-based prefixoids exist and whether they originate from the same pool of A(dv)-elements in these languages requires investigation. A cross-linguistic West Germanic comparison investigating the possible

effect of prefixoids and prefixes on the syntax and semantics of the first phase would therefore be a useful extension of the current study.

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Data availability We deposited our data in a publicly available repository, namely on the Open Science Framework platform.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Open science framework The full data set is made publicly available via Open Science Framework, and can be found here.

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References

- Acedo-Matellán, Victor, and Arkadiusz Kwapiszewski. 2024. Talmy's typology revisited: A spanning approach. *Linguistic Inquiry*.
- Ackema, Peter. 1999. The nonuniform structure of Dutch N–V compounds. In *Yearbook of morphology* 1998, ed. G. Booij and J. van Marle, 127–158. Dordrecht: Springer.
- Baayen, R. H., R. Piepenbrock, and L. Gulikers. 1995. *CELEX2 LDC96L14*. Philadelphia: Linguistic Data Consortium.
- Baker, Mark C. 2003. *Lexical categories: Verbs, nouns, and adjectives*. Cambridge: Cambridge University Press.

- Benz, Johanna. 2025. *Structure and interpretation across categories*. PhD diss: University of Pennsylvania.
- Biskup, Petr, Michael T. Putnam, and Laura Catharine Smith. 2011. German particle and prefix verbs at the syntax-morphology interface. *Leuvense Bijdragen* 97:106–135.
- Blom, Corrien. 2005. Complex Predicates in Dutch. Synchrony and diachrony. Ph.D. dissertation, LOT Dissertation Series.
- Bogomolets, Ksenia. 2020. Lexical accent in languages with complex morphology. Ph.D. dissertation, University of Connecticut.
- Booij, Geert. 1990. The boundary between morphology and syntax: Separable complex verbs in Dutch. In *Yearbook of morphology* 3, ed. Geert Booij and Jaap van Marle, 45–63. Dordrecht: Kluwer Academic Publishers.
- Cavirani-Pots, Cora. 2020. Roots in progress: Semi-lexicality in the Dutch and Afrikaans verbal domain. Ph.D. dissertation, LOT Dissertation Series.
- Cavirani-Pots, Cora, and Engela De Villiers. 2023. Univerbation in Afrikaans verbal diminutives. *Linguistics in the Netherlands* 20:23–38.
- Chomsky, Noam. 1995. *The minimalist program*. Cambridge, MA: MIT Press.
- de Haas, Wim, and Mieke Trommelen. 1993. *Morfologisch handboek van het Nederlands: Een overzicht van de woordvorming*. The Hague: SDU.
- Embick, David. 2023. Smaller syntax for English stative passives. *Acta Linguistica Academica* 70 (3): 285–316.
- Embick, David, and Rolf Noyer. 2007. Distributed morphology and the syntax/morphology interface. In *Oxford handbook of linguistic interfaces*, ed. Gillian Ramchand and Charles Reiss, 289–324. Oxford: Oxford University Press.
- Epstein, Samuel, Hisatsugu Kitahara, and T. Daniel Seely. 2016. Phase cancellation by external pair-merge of heads. *The Linguistic Review* 33 (1): 87–102.
- Fábregas, Antonio. 2020. *Morphologically derived adjectives in Spanish*. Amsterdam: John Benjamins.
- Fenger, Paula, Ava Creemers, and Marlijn Meijer. 2015. On the syntactic nature of the Dutch prefix. *Linguistics in the Netherlands* 2015 (32): 48–62.
- Folli, Raffaella, and Heidi Harley. 2020. A head movement approach to Talmy's typology. *Linguistic Inquiry* 51 (3): 425–470.
- Haeseryn, W., K. Romijn, G. Geerts, J. de Rooij, and M. C. van den Toorn. 1997. *Algemene Nederlandse Spraakkunst*, 2nd ed. Groningen: Martinus Nijhoff.
- Halle, Morris, and Alec Marantz. 1993. Distributed morphology and the pieces of inflection. In *The view from building 20*, ed. K. Hale and S. Jay Keyser, 111–176. Cambridge, MA: MIT Press.
- Harley, Heidi. 2008. The bipartite structure of verbs cross-linguistically, or Why Mary can't exhibit John her paintings. In *Conferências do V Congresso Internacional da Associação Brasileira de Linguística*, eds. T. Cristóvão Silva and H. Mello, 45–84. Belo Horizonte: ABRALIN and FAL/UFMG.
- Hoekstra, T. A., M. Lansu, and M. Westerduin. 1987. Complexe Verba. *GLoT* 10:61–77.
- Julien, Marit. 2002. *Syntactic heads and word formation*. Oxford: Oxford University Press.
- Koopman, Hilda. 1995. On verbs that fail to undergo v-second. *Linguistic Inquiry* 26:139–163.
- Kouneli, Maria, Paula Fenger, and Jonathan D. Bobaljik. 2024. Syntactic limits on phonological dominance. In *Proceedings of the 40th West Coast Conference on Formal Linguistics (WCCFL)*, eds. Jiayi Lu, Erika Petersen, Anissa Zaitso, and Boris Harizanov, 173–181. Somerville, MA: Cascadia Press.
- Lohndal, Terje. 2014. *Phrase structure and argument structure: A case study of the syntax-semantics interface*. Oxford: Oxford University Press.
- Manzini, Maria Rita, and Diego Pescarini. 2022. The clitic string as a Pair Merge sequence. *Glossa: A Journal of General Linguistics* 7 (1): 1–37.
- Marantz, Alec. 2001. Words and things. Unpublished manuscript.
- Matushanky, Ora. 2006. Head movement in linguistic theory. *Linguistic Inquiry* 37:69–109.
- McIntyre, Andrew. 2002. Idiosyncrasy in particle verbs. In *Verb-particle explorations*, ed. Andrew McIntyre and Nicole Dehé Ray. Jackendoff and Silke Urban, 95–118. Berlin: Mouton de Gruyter.
- Mitrović, Moreno, and Phoebos Panagiotidis. 2020. Adjectives exist, adjectivisers do not: A bicategorical typology. *Glossa: A Journal of General Linguistics* 5:1–28.
- Neeleman, Ad. 1994. Complex predicates. Ph.D. dissertation, Utrecht University.
- Neeleman, Ad., and Joleen Schipper. 1993. Verbal prefixation in Dutch: Thematic evidence for conversion. In *Yearbook of morphology* 1992, ed. Geert Booij and Jaap van Marle, 57–92. Amsterdam: Kluwer Academic Publishers.

- Paparounas, Lefteris. 2023a. Phrasal layering versus complex heads in Greek stative passives. In *Proceedings of 53rd annual meeting of the North East Linguistic Society* -, vol. 3, ed. Suet-Ying. Lam and Satoru Ozaki, 25–36. Amherst, MA: GLSA.
- Paparounas, Lefteris. 2023b. Voice from syntax to syncretism. Ph.D. dissertation, University of Pennsylvania.
- Ramchand, Gillian. 2008. *Verb meaning and the lexicon: A first-phase syntax*. Oxford: Oxford University Press.
- Ramchand, Gillian. 2018. *Situations and syntactic structures*. Cambridge, MA: MIT Press.
- Ramchand, Gillian. 2019. Event structure and verbal decomposition. In *The Oxford handbook of event structure*, ed. Robert Truswell, 314–341. Oxford: Oxford University Press.
- Ramchand, Gillian, and Peter Svenonius. 2002. The lexical syntax and lexical semantics of the verb-particle construction. In *Proceedings of the 21st West Coast Conference on Formal Linguistics (WCCFL)*, eds. Line Mikkelsen and Christopher Potts, 387–400. Somerville, MA: Cascadilla Press.
- Roberts, Ian. 1991. Excorporation and minimality. *Linguistic Inquiry* 22:209–218.
- Smesseart, Hans, Jeroen van Craenenbroeck, and Guido Vanden Wyngaerd. 2019. Semi-lexical categories: The function of content words and the content Basisbegrippen taalkunde. *Morfologie en Syntaxis*. Leuven: Acco.
- Svenonius, Peter. 2004. Slavic prefixes inside and outside VP. In *Nordlyd 32.2., special issue on Slavic prefixes*, 205–253. Tromsø: CASTL.
- Talmy, Leonard. 1975. Semantics and syntax of motion. In *Syntax and semantics 4*, ed. John P. Kimball, 181–238. New York: Academic Press.
- Talmy, Leonard. 1985. Lexicalization patterns: Semantic structure in lexical forms. In *Language typology and syntactic description*, vol. 3, ed. Timothy Shopen, 57–149. Cambridge: Cambridge University Press.
- Uriagereka, Juan. 1999. *Spell-out and the minimalist program*. Oxford: Oxford University Press.
- Van Huyssteen, Gerhard B. 2020. Prefixation. Retrieved May 26, 2025 from <https://taalportaal.org/taalportaal/topic/pid/topic-13998813296601200>.
- Vanden Wyngaerd, Guido, Karen De Clercq, and Pavel Caha. 2022. A nanosyntactic approach to Dutch deadjectival verbs. *Linguistics in the Netherlands* 39:240–262.
- Šereikaitė, Milena. 2017. The reanalysis of Lithuanian reflexive -si-: A DM approach. In *Proceedings of the 34th West Coast Conference on Formal Linguistics (WCCFL)*, 447–453. Somerville, MA: Cascadilla Proceedings.
- Wechsler, Stephen. 2005. Resultatives under the ‘Event -Argument Homomorphism’ model of telicity. In *The syntax of aspect: Deriving thematic and aspectual interpretation*, ed. N. Erteschik-Shir and T. Rapoport, 255–273. Oxford: Oxford University Press.
- Wood, Jim. 2023. *Icelandic nominalizations and allosemy*. Oxford: Oxford University Press.
- Wood, Jim, and Alec Marantz. 2017. The interpretation of external arguments. In *The verbal domain*, ed. Roberta D’Alessandro, Irene Franco, and Ángel. J. Gallego, 255–278. Oxford: Oxford University Press.
- Wurmbrand, Susanne. 1998. Heads or phrases? particles in particular. In *Phonology and morphology of the Germanic languages*, ed. Wolfgang Kehrein and Richard Wiese, 267–295. Tübingen: Max Niemeyer Verlag.
- Wurmbrand, Susi. 2000. The structure(s) of particle verbs. Unpublished manuscript.
- Zeijlstra, H. 2008. On the syntactic flexibility of formal features. In *The limits of syntactic variation*, ed. T. Biberauer, 143–173. Amsterdam: John Benjamins.
- Zeller, Jochen. 2001. *Particle verbs and local domains*. Amsterdam: John Benjamins.
- Zwanenburgh, Wiecher. 1997. Dutch prefixes and prepositions in complex verbs. In *Advances in morphology*, ed. Wolfgang U. Dressler, Martin Prinzhorn, and John R. Rennison, 63–77. Berlin: Mouton de Gruyter.